

Solar Inverter Monitoring Aggregates

(TimescaleDB / Tiger Cloud)

This document lists useful SQL aggregates for monitoring solar inverter systems using hypertables like `string_readings` and `inverter_readings`. Each aggregate includes a brief description of what the metric represents in a solar monitoring context.

1. Average Power Output per Inverter

Represents the average electrical power produced by an inverter during a time window. Useful for monitoring normal power generation trends and identifying underperforming inverters.

```
SELECT inverter_id, time_bucket('1 hour', time) AS bucket, AVG(voltage_v * current_a) AS avg_power FROM inverter_readings GROUP BY inverter_id, bucket;
```

2. Temperature Monitoring (Overheating Detection)

Tracks the maximum temperature reached by an inverter within a time interval. Helps detect overheating which could lead to inverter shutdown or hardware damage.

```
SELECT inverter_id, time_bucket('1 hour', time) AS bucket, MAX(temperature_c) AS max_temp FROM inverter_readings GROUP BY inverter_id, bucket;
```

3. String Performance Comparison

Measures average current for each solar string connected to an inverter. This helps compare strings and detect faulty or degraded panels.

```
SELECT inverter_id, string_number, time_bucket('1 hour', time), AVG(current_a) AS avg_current FROM string_readings GROUP BY inverter_id, string_number, time_bucket('1 hour', time);
```

4. Voltage Stability (Fluctuation Detection)

Calculates voltage variation during a time window. Large fluctuations may indicate grid instability or inverter electrical issues.

```
SELECT inverter_id, time_bucket('1 hour', time), MAX(voltage_v) - MIN(voltage_v) AS voltage_variation FROM inverter_readings GROUP BY inverter_id, time_bucket('1 hour', time);
```

5. Data Availability (Sensor Health)

Counts how many readings were received in a given time interval. This helps detect communication issues or sensor failures.

```
SELECT inverter_id, time_bucket('1 hour', time), COUNT(*) AS readings_count FROM inverter_readings GROUP BY inverter_id, time_bucket('1 hour', time);
```

```
time_bucket('1 hour', time);
```

6. Daily Energy Production

Calculates total energy produced by each inverter during a day using voltage and current measurements.

```
SELECT inverter_id, time_bucket('1 day', time), SUM(voltage_v *  
current_a) AS energy_generated FROM inverter_readings GROUP BY  
inverter_id, time_bucket('1 day', time);
```

7. Peak Power Output

Identifies the highest power output reached by an inverter during the day. Useful for evaluating maximum generation capability.

```
SELECT inverter_id, time_bucket('1 day', time) AS day, MAX(voltage_v *  
current_a) AS peak_power FROM inverter_readings GROUP BY inverter_id,  
day;
```

8. Energy Generated Per String

Measures the total energy generated by each solar string connected to an inverter.

```
SELECT inverter_id, string_number, time_bucket('1 day', time) AS day,  
SUM(voltage_v * current_a) AS string_energy FROM string_readings GROUP BY  
inverter_id, string_number, day;
```

9. Inverter Utilization Rate

Represents how much of the inverter's capacity is being used relative to its maximum possible power output.

```
SELECT inverter_id, time_bucket('1 hour', time), AVG(voltage_v *  
current_a)/5000 AS utilization_ratio FROM inverter_readings GROUP BY  
inverter_id, time_bucket('1 hour', time);
```

10. Temperature vs Power Analysis

Compares average inverter temperature with average power output to understand how overheating affects performance.

```
SELECT time_bucket('1 hour', time), AVG(temperature_c) AS avg_temp,  
AVG(voltage_v * current_a) AS avg_power FROM inverter_readings GROUP BY  
1;
```

11. Voltage Drop Detection

Detects sudden drops in voltage which may indicate electrical faults or grid issues.

```
SELECT inverter_id, time_bucket('10 minutes', time), MIN(voltage_v) AS  
min_voltage FROM inverter_readings GROUP BY inverter_id, time_bucket('10  
minutes', time);
```

12. Current Imbalance Between Strings

Measures the difference between maximum and minimum current among strings. Large imbalance suggests panel damage or shading problems.

```
SELECT inverter_id, time_bucket('1 hour', time), MAX(current_a) -  
MIN(current_a) AS current_imbalance FROM string_readings GROUP BY  
inverter_id, time_bucket('1 hour', time);
```

13. Inverter Downtime Detection

Counts instances where current output is zero, which may indicate inverter shutdowns or outages.

```
SELECT inverter_id, time_bucket('1 hour', time), SUM(CASE WHEN current_a  
= 0 THEN 1 ELSE 0 END) AS zero_output_events FROM inverter_readings GROUP  
BY inverter_id, time_bucket('1 hour', time);
```

14. Ramp Rate (Power Change Speed)

Measures how quickly power output changes over short intervals. Useful for detecting instability caused by cloud cover or system issues.

```
SELECT inverter_id, time_bucket('5 minutes', time), MAX(voltage_v *  
current_a) - MIN(voltage_v * current_a) AS power_ramp FROM  
inverter_readings GROUP BY inverter_id, time_bucket('5 minutes', time);
```

15. Panel Efficiency Trend

Tracks average daily power production to observe long-term efficiency changes in solar panels.

```
SELECT time_bucket('1 day', time), AVG(voltage_v * current_a) AS  
avg_daily_power FROM string_readings GROUP BY 1;
```

16. Night-Time Production Check

Checks whether power is being generated during night hours, which usually indicates sensor or data errors.

```
SELECT COUNT(*) FROM inverter_readings WHERE voltage_v * current_a > 0  
AND EXTRACT(HOUR FROM time) BETWEEN 0 AND 5;
```